

Section A

- 1** A bungee jumper of mass 60 kg is attached to an elastic rope which starts to stretch after a short time of free fall. The gravitational potential energy of the bungee jumper is 0 J when she has fallen through 40 m to the lowest point.

(a)

	gravitational potential energy / kJ	elastic potential energy / kJ	kinetic energy / kJ
top		0	0
half-way		2.6	
bottom	0		

Fig. 1.1

Fill up the missing energies at the top, bottom and half-way positions in Fig 1.1. Drag forces can be considered negligible.

[2]

	gravitational potential energy / kJ	elastic potential energy / kJ	kinetic energy / kJ
top	24 (23.5)	0	0
half-way	12 (11.8)	2.6	9.2 (9.17) (accept 9.4)
bottom	0	24 (23.5)	0

Fig. 1.1

Fill up the missing energies at the top, bottom and half-way positions in Fig 1.1. Drag forces can be considered negligible.

[2]

GPE (using mgh) [B1], EPE and KE at half-way and bottom positions (using conservation of energy) [B1]

Comments:

Most students are able to answer this part.

Some students seemed to forget that the total energy is the same regardless of the position.

(b)

Show that the unstretched length of the elastic rope is 10 m.

[2]

		Let the spring constant of the elastic rope be k and unstretched length be x At half-way position, $2.6 = \frac{1}{2}k(20 - x)^2$ At bottom position, $23.5 = \frac{1}{2}k(40 - x)^2$ [B1 for equations] $\frac{2.6}{23.5} = \frac{(20-x)^2}{(40-x)^2}$ [M1] $\sqrt{\frac{2.6}{23.5}} = \frac{20-x}{40-x}$ $x = 10 \text{ m}$ [A0] Comments: Many students are not able to form up the 2 equations.	
	(c)	Determine at what extension will the kinetic energy of the bungee jumper be the highest.	
		extension =	[3]
		m	
		Initially, when the bungee jumper jumps, her weight is greater than the tension in the elastic rope. When the tension increases until it is equal to the weight, she stops accelerating and reaches her highest velocity and kinetic energy. $mg = ke$ where e is the extension when her speed is the highest [B1] $\frac{1}{2}kx^2 = \frac{1}{2}k(30)^2 = 23500$ $k = 52.2$ [C1] $e = \frac{60 \times 9.81}{52.2} = 11.3 \text{ m}$ [A1] Comments: Most students are unable to identify the location of max velocity as the location where there is no resultant force.	
			[Total: 7]